

Department of Computer Science and Engineering

Course Code: CSE222	Credits: 1.5
Course Name: Object Oriented Programming Lab	Faculty: FRS

Lab 02 – Basics of Array and Inheritance

✓ **Task 1:** Find the maximum and minimum elements in an array.

Sample Input: [5, 3, 9, 2, 1, 7]

Output: Maximum: 9

Minimum: 1

✓ **Task 2:** Reverse the elements of an array. Store it in a new one and print the new Array. The size of the array is not specified, means it could be 5 or 7 or any integer value.

Sample Input: [1, 2, 3, 4, 5]

Output: [5, 4, 3, 2, 1]

Sample Input: [5, 3, 9, 2, 1, 7]

Output: [7, 1, 2, 9, 3, 5]

✓ **Task 3:** Remove duplicates from a sorted array.

Sample Input: [1, 1, 2, 2, 3, 4, 5, 5]

Output: [1, 2, 3, 4, 5]

✓ **Task 4:** Given an array containing integers from 1 to n with one element missing, find the missing number.

Sample Input: [1, 2, 4, 5, 6]

Output: Missing Number: 3

Task 5: Given an array of integers and a target sum, find two numbers in the array that add up to the target sum.

Sample Input: [2, 7, 11, 15], 9

Output: 2, 7

Task 6: Given an array of integers, move all the zeroes to the end of the array while maintaining the relative order of non-zero elements.

Sample Input: [0, 1, 0, 3, 12]

Output: [1, 3, 12, 0, 0]

✓ **Task 7:** Complete the **Animal**, **Dog**, **Cat** classes. So that it matches the output.

```
public class AnimalInheritance {  
    public static void main(String[] args) {  
        Animal animal = new Animal("Generic Animal");  
        Dog dog = new Dog("Buddy");  
        Cat cat = new Cat("Whiskers");  
  
        animal.speak();  
        dog.speak();  
        cat.speak();  
    }  
}
```

Output:

An animal makes a sound.

Buddy barks.

Whiskers meows.

 **Task 8:** Complete the **Shape**, **Circle**, **Rectangle** classes. So that it matches the output.

```
public class ShapeHierarchy {  
    public static void main(String[] args) {  
        Circle circle = new Circle(5.0);  
        Rectangle rectangle = new Rectangle(4.0, 6.0);  
  
        circle.displayType();  
        rectangle.displayType();  
    }  
}
```

Output:

Type of shape: Circle

Type of shape: Rectangle

Task 9: Complete the **Employee**, **Manager**, **Developer** classes. So that it matches the output.

```
public class EmployeeHierarchy {  
    public static void main(String[] args) {  
        Manager manager = new Manager("Alice", "HR");  
        Developer developer = new Developer("Bob", "Java");  
  
        manager.displayInfo();  
        developer.displayInfo();  
    }  
}
```

Output:

Name: Alice

Department: HR

Name: Bob

Programming Language: Java

Task 10: Complete the **Account**, **SavingsAccount**, **CheckingAccount** classes. So that it matches the output.

```
public class BankAccountHierarchy {  
    public static void main(String[] args) {  
        SavingsAccount savingsAccount = new SavingsAccount("SA001",  
1000.0, 0.05);  
        CheckingAccount checkingAccount = new CheckingAccount("CA001",  
500.0, 200.0);  
  
        savingsAccount.displayBalance();  
        savingsAccount.addInterest();  
        savingsAccount.displayBalance();  
  
        checkingAccount.displayBalance();  
        checkingAccount.withdraw(700.0);  
        checkingAccount.displayBalance();  
    }  
}
```

Output:

Account SA001 balance: 1000.0

Interest added to account SA001

Account SA001 balance: 1050.0

Account CA001 balance: 500.0

Withdrew 700.0 from checking account CA001

Account CA001 balance: -200.0